Amrita School of Artificial Intelligence, Faridabad
Amrita Vishwa Vidyapeetham

Course Code: **23MAT214**

Mathematics for Intelligent System-4

Part- 2.1

Matrix Splitting and Proximal Algorithms

BTech 4th Sem

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*All Sources: Stanford University, Texas Tech University, UC Santa Barbara, etc.
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MATRIX SPLITTING

Solve a linear system: $Ax = b$
where:
- $A \in \mathbb{R}^{n \times n}$
- $x, b \in \mathbb{R}^{n}$

We decompose:

$$A = M - N$$

where:
- M: easily invertible matrix
- N: remaining part

Starting from:

$$Ax = b \Rightarrow (M - N)x = b$$

Rearrange:

$$Mx = Nx + b$$

Multiply by M^{-1}:

$$x = M^{-1}Nx + M^{-1}b$$

Iterative update:

$$x^{(k+1)} = M^{-1}Nx^{(k)} + M^{-1}b$$

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(A) Jacobi Method

Solve a linear system: $Ax = b$
where:
- $A \in \mathbb{R}^{n \times n}$
- $x, b \in \mathbb{R}^{n}$

Start:

$$Ax = b$$

Split:

$$A = D + (L + U)$$

Diagonal part D
Lower triangular part L
Upper triangular part U

Substitute:

$$\big(D + (L + U)\big)x = b$$

Rearrange:

$$Dx = b - (L + U)x$$

Multiply by D^{-1}:

$$x = D^{-1}b - D^{-1}(L + U)x$$

Replace x on RHS with previous value:

$$\boldsymbol{x^{(k+1)} = D^{-1}b - D^{-1}(L + U)x^{(k)}}$$

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NUMERICAL EXAMPLE

Solve:

$$Ax = b$$

Take a concrete system:

$$\begin{cases} 4x_1 + x_2 = 9 \\ 2x_1 + 3x_2 = 13 \end{cases}$$

Matrix form:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, b = \begin{bmatrix} 9 \\ 13 \end{bmatrix}$$

Split: $A = D + (L + U)$

Diagonal part D
Lower triangular part L
Upper triangular part U

Apply to your matrix:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Then:

$$D = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix} \quad L = \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} \quad U = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

check:

$$D + L + U$$

$$= \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

NUMERICAL EXAMPLE

Solve:

$$Ax = b$$

Take a concrete system:

$$\begin{cases} 4x_1 + x_2 = 9 \\ 2x_1 + 3x_2 = 13 \end{cases}$$

Matrix form:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, b = \begin{bmatrix} 9 \\ 13 \end{bmatrix}$$

Iteration formula:

$$\boldsymbol{x^{(k+1)} = D^{-1}\left(b - (L+U)x^{(k)}\right)}$$

Split matrix

$$A = D + (L + U)$$

$$D = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}, L + U = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$$

$$D^{-1} = \begin{bmatrix} 1/4 & 0 \\ 0 & 1/3 \end{bmatrix}$$

Initial guess Let: $x^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Iteration 1

Compute:

$$(L + U)x^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$b - (L + U)x^{(0)} = \begin{bmatrix} 9 \\ 13 \end{bmatrix}$$

Apply D^{-1}:

$$x^{(1)} = \boldsymbol{D^{-1}} * \begin{bmatrix} 9 \\ 13 \end{bmatrix} = \begin{bmatrix} 9/4 \\ 13/3 \end{bmatrix} = \begin{bmatrix} 2.25 \\ 4.33 \end{bmatrix}$$

NUMERICAL EXAMPLE

Solve:

$$Ax = b$$

Take a concrete system:

$$\begin{cases} 4x_1 + x_2 = 9 \\ 2x_1 + 3x_2 = 13 \end{cases}$$

Matrix form:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, b = \begin{bmatrix} 9 \\ 13 \end{bmatrix}$$

Iteration formula:

$$\boldsymbol{x^{(k+1)} = D^{-1}\left(b - (L+U)x^{(k)}\right)}$$

$$D = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}, L + U = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$$

Iteration 1

$$x^{(1)} = \begin{bmatrix} 9/4 \\ 13/3 \end{bmatrix} = \begin{bmatrix} 2.25 \\ 4.33 \end{bmatrix}$$

Iteration 2

Compute:

$$(L+U)x^{(1)} = \begin{bmatrix} 4.33 \\ 4.5 \end{bmatrix}$$

$$b - (L+U)x^{(1)} = \begin{bmatrix} 9 - 4.33 \\ 13 - 4.5 \end{bmatrix} = \begin{bmatrix} 4.67 \\ 8.5 \end{bmatrix}$$

$$x^{(2)} = \begin{bmatrix} 4.67/4 \\ 8.5/3 \end{bmatrix} = \begin{bmatrix} 1.17 \\ 2.83 \end{bmatrix}$$

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Iteration (k)	$x_1^{(k)}$	$x_2^{(k)}$
0	0.000	0.000
1	2.250	4.333
2	1.167	2.833
3	1.542	3.889
4	1.278	3.306
5	1.423	3.481
6	1.380	3.384
7	1.404	3.413
8	1.397	3.397
9	1.401	3.402
10	1.399	3.399

(B) Gauss-Seidel Method

Start:

$$Ax = b$$

Now we choose:

$$M = D + L$$
$$N = -U$$

Substitute into equation

$$\big((D + L) + U\big)x = b$$
$$(D + L)x = b - Ux$$

Iterative form: Replace RHS x with previous iteration:

$$(D + L)x^{(k+1)} = b - Ux^{(k)}$$

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NUMERICAL EXAMPLE

Solve:

$$Ax = b$$

Take a concrete system:

$$\begin{cases} 4x_1 + x_2 = 9 \\ 2x_1 + 3x_2 = 13 \end{cases}$$

Matrix form:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, b = \begin{bmatrix} 9 \\ 13 \end{bmatrix}$$

Iterative form:

$$\boldsymbol{(D + L)x^{(k+1)} = b - Ux^{(k)}}$$

Split matrices $D + L = \begin{bmatrix} 4 & 0 \\ 2 & 3 \end{bmatrix}, U = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

Compute RHS

Let: $x^{(k)} = \begin{bmatrix} x_1^{(k)} \\ x_2^{(k)} \end{bmatrix}$ $Then$ $Ux^{(k)} = \begin{bmatrix} x_2^{(k)} \\ 0 \end{bmatrix}$

So: $b - Ux^{(k)} = \begin{bmatrix} 9 - x_2^{(k)} \\ 13 \end{bmatrix}$

Solve triangular system

$$\begin{bmatrix} 4 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1^{(k+1)} \\ x_2^{(k+1)} \end{bmatrix} = \begin{bmatrix} 9 - x_2^{(k)} \\ 13 \end{bmatrix}$$

Step 5: Solve row by row

Row 1:

$$4x_1^{(k+1)} = 9 - x_2^{(k)}$$

$$x_1^{(k+1)} = \frac{1}{4}\left(9 - x_2^{(k)}\right)$$

Row 2:

$$2x_1^{(k+1)} + 3x_2^{(k+1)} = 13$$

$$x_2^{(k+1)} = \frac{1}{3}\left(13 - 2x_1^{(k+1)}\right)$$

NUMERICAL EXAMPLE

Solve:

$$Ax = b$$

Take a concrete system:

$$\begin{cases} 4x_1 + x_2 = 9 \\ 2x_1 + 3x_2 = 13 \end{cases}$$

Matrix form:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, b = \begin{bmatrix} 9 \\ 13 \end{bmatrix}$$

Iterative form:

$$\boldsymbol{(D + L)x^{(k+1)} = b - Ux^{(k)}}$$

Split matrices $D + L = \begin{bmatrix} 4 & 0 \\ 2 & 3 \end{bmatrix}, U = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

So: $$b - Ux^{(k)} = \begin{bmatrix} 9 - x_2^{(k)} \\ 13 \end{bmatrix}$$

Solve triangular system

$$\begin{bmatrix} 4 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1^{(k+1)} \\ x_2^{(k+1)} \end{bmatrix} = \begin{bmatrix} 9 - x_2^{(k)} \\ 13 \end{bmatrix}$$

Solve row by row

Row 1:

$$4x_1^{(k+1)} = 9 - x_2^{(k)}$$

$$x_1^{(k+1)} = \frac{1}{4}\left(9 - x_2^{(k)}\right)$$

Row 2:

$$2x_1^{(k+1)} + 3x_2^{(k+1)} = 13$$

$$x_2^{(k+1)} = \frac{1}{3}\left(13 - 2x_1^{(k+1)}\right)$$

NUMERICAL EXAMPLE

Solve:

$$Ax = b$$

Take a concrete system:

$$\begin{cases} 4x_1 + x_2 = 9 \\ 2x_1 + 3x_2 = 13 \end{cases}$$

Matrix form:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, b = \begin{bmatrix} 9 \\ 13 \end{bmatrix}$$

Iterative form:

$$(\boldsymbol{D} + \boldsymbol{L})\boldsymbol{x}^{(k+1)} = \boldsymbol{b} - \boldsymbol{U}\boldsymbol{x}^{(k)}$$

Solve

$$x_1^{(k+1)} = \frac{1}{4}\left(9 - x_2^{(k)}\right)$$

$$x_2^{(k+1)} = \frac{1}{3}\left(13 - 2x_1^{(k+1)}\right)$$

Iteration 1

Start:

$$x^{(0)} = (0, 0)$$

$$x_1^{(1)} = 9/4 = 2.25$$

$$x_2^{(1)} = (13 - 2 \times 2.25)/3 = 8.5/3 = 2.83$$

Iteration 2

$$x_1^{(2)} = (9 - 2.83)/4 = 1.54$$

$$x_2^{(2)} = (13 - 2 \times 1.54)/3 = 3.64$$

👉 Converges faster than Jacobi

Iteration (k)	$x_1^{(k)}$	$x_2^{(k)}$
0	0.000	0.000
1	2.250	2.833
2	1.542	3.306
3	1.423	3.384
4	1.404	3.397
5	1.401	3.399
6	1.400	3.400

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PROXIMAL ALGORITHMS

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We want to solve:

$$\min_x f(x) + g(x)$$

where:

$f(x)$:smooth (differentiable)

$g(x)$:possibly **non-smooth**

Problems:

- Large-scale systems
- Non-smooth terms
- Direct inversion is expensive

👉 So we **convert one hard problem → many simple steps**

Example:

$$g(x) = |x|$$

👉 Not differentiable at $x = 0$

So:

- Gradient descent ❌ fails
- Classical calculus ❌ fails

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